

Mouchette + Arachne Five Strike / Mortal Blast pairs

97,228

Combined damage before target mitigation, for the approved setup below. The opening Exalt attack is not included. This report replaces all earlier PDFs of this five-pair example.

Contribution	Damage
Mortal Blast: 5 casts, 2 hits each	42,230
Dramatic Encounter: 4 pursuits, 3 hits each	46,458
Mouchette subtotal	88,688
Arachne: 5 Strikes	8,540
Combined	97,228

The setup you approved

Both Awakeners are level 90, with maximum skills (level 6) and Gnostic Potential (level 5). Mouchette is E2 and Soulforge 10; she Rouses before Exalt. Arachne is unroused. Both have 100% crit rate and a 150% Crit DMG stat. Starting STR is 0.

Astral Reign; one Dominator; no additional modifiers. After Rouse and Exalt, play **[Arachne Strike -> Mortal Blast] five times in the same turn**. Start with zero pursuit progress and zero personal Strike stacks. Let each pursuit resolve before playing the next pair.

The question supplies five casts: sufficient cards and energy are treated as available. No separate Arachne Soulforge, Rouse/Fate Cut, equipment, relic, extra target state or other damage source is added. The target must survive the complete sequence.

What this number means

The sourced setup and recovered formula produce the total above. Fourteen resolved per-hit records were rechecked through original Lua arithmetic and matched exactly. This is **not an independently verified replay or a complete battle simulation**. Shield, immunity, boss caps, phases or other target mechanics would change actual HP loss.

Inputs and stacking order

SKeyDB stat calculation [S1-S3]

Max Gnostic Potential adds 10 base-attribute levels. SKeyDB rounds the base stat up, then rounds the Soulforge percentage result up. Its stat calculation uses a 0.000000001 floating-point tolerance.

Quantity	Calculation	Result
Mouchette base ATK	$\text{ceil}((20 + 90 + 10) \times 1.65)$	198
Mouchette ATK, Soulforge 10	$\text{ceil}(198 \times 1.30)$	258
Arachne ATK	$\text{ceil}((20 + 90 + 10) \times 1.15)$	138
Exalt flat Strike contribution	$\text{ceil}(258 \times 250\%)$	645
Mortal Blast / pursuit argument	$\text{ceil}(258 \times 30\%)$	78
Arachne Strike argument	$\text{ceil}(138 \times 20\%)$	28

Separate multipliers, separate owners [P1-P3]

- 1. Ordinary skill argument:** ceil the ATK coefficient before it becomes stored Arg1. Explicit card argument overrides are a different path; none are supplied here.
- 2. Soulforge Strike Base DMG:** +35%, or $\times 1.35$, scales the base argument for both Awakeners. It does not multiply the separate +645 Exalt contribution.
- 3. Flat Exalt bonus:** add 645 to each eligible Strike hit. STR contributes 0. The flat bonus applies once per hit, not once per card.
- 4. Personal Strike stacks:** Mouchette multiplies that subtotal by $(1 + 0.25 \times k)$. Arachne does not inherit this personal multiplier. Here k is the number of completed pursuits.
- 5. Crit and Dominator:** a 150% Crit DMG stat adds 150% to normal damage: $\times 2.5$, not $\times 1.5$. Mouchette alone then receives the Soulforge +50% Dominator factor, $\times 1.5$. Arachne receives no such bonus in this setup.

Per-hit formulas for this baseline

Mouchette: $U(k) = \text{ceil}(((78 \times 1.35) + 645) \times (1 + 0.25k) - 0.000001)$.

$D(k) = \text{ceil}(U(k) \times 2.5 \times 1.5)$.

Arachne: $U = \text{ceil}((28 \times 1.35) + 645 - 0.000001) = 683$.

$D = \text{ceil}(683 \times 2.5) = \mathbf{1,708 \text{ per Strike}}$.

The recovered utility also applies its later floor/minimum stage; with neutral reductions and these positive integer outputs, it leaves U unchanged. Target rounding is performed per hit before summing hits.

All five pairs, step by step

Roused Mouchette has three pursuit hits. E2 Mortal Blast has two hits; the E3 Exalt extra-hit upgrade is absent. Every two played Strikes cause one pursuit, capped at four per turn. Mortal Blast counts as a Strike; the pursuit does not recursively count as another played Strike. [S4, P2]

Pair	k before	Mouchette U	Mouchette / hit	Blast x2	Pursuit x3	Arachne	Pair total
1	0	751	2,817	5,634	8,451	1,708	15,793
2	1	938	3,518	7,036	10,554	1,708	19,298
3	2	1126	4,223	8,446	12,669	1,708	22,823
4	3	1314	4,928	9,856	14,784	1,708	26,348
5	4	1501	5,629	11,258	0	1,708	12,966

Pair 1: no personal stack yet

The base-plus-flat subtotal is $(78 \times 1.35) + 645 = 750.3$. With $k = 0$, the utility rounds this to 751. Crit and Dominator give $\text{ceil}(751 \times 2.5 \times 1.5) = \text{ceil}(2,816.25) = \mathbf{2,817}$ per Mouchette hit.

Arachne contributes 1,708. Mortal Blast contributes $2 \times 2,817 = 5,634$. The pursuit contributes $3 \times 2,817 = 8,451$. Pair 1 total: **15,793**. Only after the pursuit damage does Mouchette gain her first +25% personal Strike layer.

Pairs 2-4: the pursuit adds the next stack afterward

At $k = 1, 2$ and 3 , the utility subtotals are 937.875, 1,125.45 and 1,313.025. They round to 938, 1,126 and 1,314. The final per-hit amounts are 3,518, 4,223 and 4,928. Each pursuit uses its current k for all three hits, then adds one layer. [P2]

Pair 5: four layers, but the pursuit cap is reached

At $k = 4$, $750.3 \times 2 = 1,500.6$, rounded to 1,501. Each Blast hit is $\text{ceil}(1,501 \times 2.5 \times 1.5) = 5,629$. There is no fifth pursuit: $1,708 + 2 \times 5,629 = \mathbf{12,966}$.

Final addition

Mortal Blast: $2 \times (2,817 + 3,518 + 4,223 + 4,928 + 5,629) = 42,230$.

Pursuits: $3 \times (2,817 + 3,518 + 4,223 + 4,928) = 46,458$.

Arachne: $5 \times 1,708 = 8,540$.

Total: $42,230 + 46,458 + 8,540 = \mathbf{97,228}$.

Evidence and reproducibility

SKeyDB: character data and coefficients

[S1] Stat scaling

[S2] Mouchette

[S2] Arachne

[S3] Gnostic Potential

[S3] Soulforge Aptitude

[S4] Mortal Blast

[S4] Shining Tornado

[S4] Arachne Strike

[S4] Dramatic Encounter

Pinned commit: a2ff07765b2e0c78af827577f881054dbe224d8d. Character data is credited to dansa / SKeyDB. This report adapts its data and adds calculations and recovered-code analysis. Data-derived material is shared under [CC BY-NC-SA 4.0](#); see [SKeyDB data license](#).

Recovered PC code and configuration

[P1] Arithmetic: BattleCmdServer.GetSkillArgs, __GetShowDamage and __GetFinalDamage; BattleUtilServer.ShowDamageFormula. Ordinary argument ceiling, flat/percentage order, critical multiplier and final ceiling were recovered from PC resource 144/build 51.

[P2] Sequence: Skill 122483 (Blast), Skill 123159 (pursuit), Cmd 123160 (two-Strike trigger), Cmd 123163 (pursuit damage before personal layer), State 123165 (four-pursuit counter), State 123168 (+25% inside Strike per layer). Cmd 122491 and State 124039 establish that the extra hit requires E3. BEAfterUseCard suppresses the pursuit's recursive played-card event.

[P3] Soulforge and Exalt: Talent 122481 level 10, State 123703 (+35% outside Strike), State 124656 (+50% damage_per2monster_boss), State 124034 (+645 strikecard_damage_plus here). Player-owned flat Strike contributions are distributed to Awakeners; they must not be counted twice.

What was checked, and what was not

The reproducible JSON includes input vectors, per-hit traces, source SHA-256 hashes and 14 matching original-runtime arithmetic checks. The generator is tools/mouchette-approved-baseline.mjs; the checker is tools/verify_mouchette_baseline.py. The accompanying JSON is output/pdf/mouchette-approved-baseline-calculation.json.

These checks exercise arithmetic with resolved inputs. They do not independently establish a whole-battle event trace, target survival or a particular boss's caps/phases. The sequence is a conditional worked example using the recovered configuration. It does not satisfy the calculator's gameplay publication gate.